

and for faster development. Companies are using PaaS platforms like Heroku to reduce infrastructure investments. Luckily, RoR supports all the major PaaS platforms. It also incorporates RESTful architecture. RoR handles RESTful APIs properly, so developers can focus on the main logic rather than the API build-up.

#### ■ It follows the MVC design pattern

RoR follows the Model View Controller (MVC) design where the model can be co-related with database-side coding, the view is for Web pages/HTML templates, and the controller acts as a mediator between the front-end and the back-end.

#### ■ Strong community support

Ruby on Rails has very strong community support. Whenever beginners join the RoR community, they get plenty of help from everyone. Along with that, RoR has a very rich set of free plugins, which are readymade functionalities and are ready to use. Rather than building from scratch, developers can use these plugins to speed up their development process.

#### ■ Object-oriented

In RoR, everything is an object. It is highly object-oriented and easy to learn. Even numbers and strings are objects.

## Installation steps

**Windows:** To install Ruby on Rails, we need to install components like Ruby, Rails, Git, SQLite and some of the other components that Rails needs to be able to run on Windows. It is a cumbersome task, but *railsinstaller* has made our lives easier. We have to go to <http://railsinstaller.org/> and just click on *Installer*. Currently, the latest version is *railsinstaller-3.2.0.exe*.

Mac users can also use this URL as a reference to install Ruby on Rails.

Figure 1 demonstrates the page from where we can download *railsinstaller*, which will automatically install all the components to run Ruby on Rails.

After installing Rails, try to run `rails -v` to check the version of Rails installed. Sometimes, you may get an error like the one shown below:

*"The system cannot find the path specified."*

This is because some *.bat* files have the wrong path.

To correct this, simply open your favourite text editor—in my case it's the Sublime text editor—and take the following steps.

Open all the files under `C:\RailsInstaller\Ruby2.2.0\bin` in the text editor and use the *Find and replace* option (the shortcut key for the Sublime text editor is: `Ctrl+Shift+F`). Now, replace `@ C:\Users\emachnic\GitRepos\railsinstaller-windows\stage\Ruby2.2.0\bin\ruby.exe` with `@%~dp0ruby.exe`.

Once this is done, you can run the `rails -v` command and



Figure 1: Ruby on Rails: Windows installation

you should not get any error message.

**Linux (Ubuntu):** Resolving software dependencies, installation and maintaining software versions successfully sometimes becomes difficult in Linux based distributions. But, RVM (Ruby Version Manager) has made our lives easy while installing Ruby on Rails.

Ubuntu is the most popular Linux distribution available in the market, so considering that, we have demonstrated the installation steps in Ubuntu OS.

To install Ruby on Rails with RVM, run the following commands as a normal user:

```
gpg --keyserver hkp://keys.gnupg.net --recv-keys
409B6B1796C275462A1703113804BB82D39DC0E3 & curl -sSL https://
get.rvm.io | bash -s stable --rails
```

Enter the normal user password to proceed further. Now, let's see how this command works.

The `gpg` command is used to connect with a public key server and to request a key attached with that particular ID. Here, we are fetching the RVM project's key, which is utilised to sign every RVM release. Having the RVM project's public key permits us to check the legitimacy of the release, which we will download now and sign with the matching private key.

The remaining portion of the command, starting with `curl`, uses the `Curl` utility to grab a file from the RVM website. It is a script that will automatically install all related components. Here `"\` is not an alias for any command.

The `-s` flag is for silent mode, and the `-S` flag overrides it to display errors if `Curl` fails. The `-L` flag is for redirection.

The script is then channelised to the bash prompt to run. The `-s` flag indicates that the input is coming from *standard in*. The rest of the commands are used to install the latest and most stable version of Ruby and Rails, along with Ruby.

After this process, we have to source the RVM scripts by using the command below:

```
source ~/.rvm/scripts/rvm
```

The Ruby on Rails framework is now installed and we can start building a Web application. To install a specific version